

CHAPTER 1

INTRODUCTION

1.1 Background

The global burden of diet-related diseases has reached unprecedented levels. According to the World Health Organization (WHO), unhealthy diets are among the top risk factors for non-communicable diseases (NCDs), which collectively account for approximately 71% of all deaths globally. In India, the National Family Health Survey (NFHS-5, 2019–2021) reports that 24% of adults are overweight or obese, while 18.7% of women and 16.2% of men are underweight — highlighting a dual burden of malnutrition.

Despite growing health consciousness in the post-COVID era, traditional methods of dietary tracking remain highly challenging. Manual food logging — manually searching databases, reading nutritional labels, or maintaining food diaries — is tedious, error-prone, and unsustainable. Research published in the *Journal of Medical Internet Research* indicates that manual food logging has an abandonment rate exceeding 60% within the first two weeks due to logging fatigue. Furthermore, generic calorie-tracking apps fail to provide personalized recommendations that account for individual medical conditions, blood work results, or daily dietary context.

Recent advancements in Artificial Intelligence, particularly in computer vision and Large Language Models (LLMs), present transformative opportunities to address these issues. Vision-capable models such as GPT-4o can now identify food items from photographs with remarkable accuracy, while conversational AI can provide personalized dietary guidance that was previously accessible only through expensive professional consultations. **MealMentor AI** is conceived at this intersection — an intelligent Android application combining AI-powered food recognition, automated nutritional analysis, comprehensive meal tracking, interactive data visualization, and a conversational AI dietitian.

1.2 Objectives

1. To design and develop an Android application that uses AI-powered computer vision (Azure OpenAI GPT-4o) to identify food items from photographs and extract detailed nutritional information.
2. To implement a comprehensive daily meal tracking system with persistent food logging and progress monitoring against personalized targets.
3. To build an AI Dietitian chat module — a context-aware conversational assistant providing personalized meal recommendations based on health profile, medical conditions, and daily food intake.
4. To create interactive data visualizations (line charts, donut charts, bar graphs, radar charts) presenting nutritional trends over daily, weekly, and monthly periods.
5. To implement health report analysis that allows users to upload medical reports for AI-based parsing, integrating extracted health markers into dietary recommendations.
6. To deploy the backend on Microsoft Azure cloud services, demonstrating proficiency in cloud-native application development.

1.3 Purpose, Scope, and Applicability

1.3.1 Purpose

The primary purpose of MealMentor AI is to bridge the accessibility gap in personalized nutrition guidance. Professional dietitian consultations cost ₹500–₹2,000 per session in India, making them financially inaccessible to a significant portion of the population. By providing a 24/7 AI-powered dietitian, MealMentor AI democratizes access to tailored dietary advice. Additionally, it aims to reduce manual logging fatigue through image-based food identification, eliminate inaccurate self-reporting, and connect isolated health data (such as blood work) directly to daily meal choices.

1.3.2 Scope

In Scope:

- Android mobile application using React Native (Expo)
- AI-powered food image recognition and nutritional analysis
- User authentication and health profile management
- Daily meal logging (AI-assisted and manual entry)
- Interactive nutritional charts and analytics dashboard
- Conversational AI Dietitian with context-aware recommendations
- Health report upload and AI-based parsing
- Cloud deployment on Microsoft Azure

Out of Scope:

- iOS application (future extension via React Native cross-platform)
- Wearable device integration
- Barcode/QR code scanning for packaged foods
- Multi-language support (English only in current version)

1.3.3 Applicability

MealMentor AI is highly applicable in various real-world scenarios:

1. **Personal Health Tracking:** Individuals seeking to maintain a healthy lifestyle, lose weight, or track macronutrients and micronutrients efficiently.
2. **Chronic Disease Management:** Patients managing diet-sensitive conditions such as diabetes, hypertension, obesity, or cardiovascular diseases can align their meals with clinical targets.
3. **Clinical Follow-ups:** Healthcare professionals can use the parsed reports and meal history to monitor patients' compliance with dietary recommendations.
4. **Corporate Wellness:** Organizations can deploy the platform to improve employee health outcomes by promoting healthy eating habits.

1.4 Achievements

The system aims to achieve several key milestones:

1. **Frictionless Food Logging:** Reducing food logging time by over 80% through one-tap image recognition instead of database searching.
2. **High Recognition Accuracy:** Achieving at least 85% accuracy in identifying food items and portion sizes from images.
3. **Hyper-Personalized Guidance:** Delivering context-aware meal recommendations that dynamically adjust based on user profile, blood reports, and current daily intake.
4. **Scalable Cloud Infrastructure:** Establishing a robust, secure, and cost-effective backend architecture using Microsoft Azure services.

1.5 Organization of Report

The project report is organized as follows:

- **Chapter 1: Introduction** – Outlines the project background, objectives, purpose, scope, applicability, achievements, and report structure.
- **Chapter 2: Problem Statement** – Identifies key challenges in current dietary management systems and outlines how MealMentor AI addresses them.
- **Chapter 3: Literature Survey** – Reviews existing applications and research papers in food recognition and LLMs, highlighting the research gap.
- **Chapter 4: Proposed System** – Explains the system overview, key features, and details the conversational AI dietitian module.
- **Chapter 5: System Architecture** – Illustrates the high-level architecture, key data flows, and database design.
- **Chapter 6: Technology Stack** – Details the frontend, backend, and cloud services utilized, along with hardware and software requirements.
- **Chapter 7: Implementation Plan & Timeline** – Presents the 20-week phased schedule and Gantt chart.
- **Chapter 8: Expected Outcomes & Future Scope** – Summarizes expected results and details future project extensions.
- **Chapter 9: References** – Lists academic papers, documentation, and resources cited.